

*Part A***Question 1:**

Let R be the ring,

$$\mathbb{Z}[\sqrt{-14}]$$

a)

Let A be the subring,

$$A = \{a + b\sqrt{-14} : a, b \in \mathbb{Z}, a - b \equiv 0 \pmod{5}\}$$

As $a \equiv b \pmod{5}$ there exists some $k \in \mathbb{Z}$ such that $a = b + 5k$.

Hence,

$$\begin{aligned} a + b\sqrt{-14} &= b + 5k + b\sqrt{-14} \\ &= b(1 + \sqrt{-14}) + 5k \end{aligned}$$

So

$$A = \langle 1 + \sqrt{-14}, 5 \rangle$$

By Definition 2.1 HB chapter 19 p94, and $R = \mathbb{Z}[\sqrt{-14}] = \langle 1, \sqrt{-14} \rangle_{\mathbb{Z}}$, so we only need to check multiplication by $\sqrt{-14}$, as 1 is trivial.

$$\sqrt{-14}(1 + \sqrt{-14}) = \sqrt{-14} + (\sqrt{-14})^2$$

and since

$$-14 \equiv 1 \pmod{5}$$

and

$$-14 = 1 - 15$$

So

$$\begin{aligned} \sqrt{-14} - 14 &= (\sqrt{-14} + 1) - 15 \\ &= (\sqrt{-14} + 1) + 5(-3) \end{aligned}$$

As $\sqrt{-14} \in A$ and $15 = 3(5) \in A$, it follows that A is closed under multiplication by any element of R , and hence A is an ideal of R .

b)

Question 2:

$$f(x) = 3x^4 + 156x^3 + 26x^2 + 130x + 26 \in \mathbb{Z}[x]$$

a)

Using Theorem 4.15 HB Chapter 12 p52,

using the prime $p = 13$

- $13 \mid 156$
- $13 \mid 26$
- $13 \mid 130$
- $13 \mid 26$
- $13 \nmid 3$
- $13^2 \nmid 26$

Hence by Eisenstein's criterion, $f(x)$ is irreducible over $\mathbb{Q}[x]$.**b)**

Let,

$$\alpha \in \mathbb{C} \quad \beta \in \mathbb{C}$$

be two distinct roots of $f(x)$

Since

$$f(x) \in \mathbb{Z}[x] \subset \mathbb{Q}[x]$$

is irreducible over $\mathbb{Q}$, and $\alpha \in \mathbb{C}$ satisfies $f(\alpha) = 0$ it follows that the minimal polynomial of α over $\mathbb{Q}$ is $f(x)$.

Using Theorem 2.21 HB Chapter 18 p105,

$$\mathbb{Q}(\alpha) \cong \mathbb{Q}[x]/(f(x))$$

The same applies to β , and hence

$$\mathbb{Q}(\beta) \cong \mathbb{Q}[x]/(f(x))$$

Thus,

$$\mathbb{Q}(\alpha) \cong \mathbb{Q}(\beta)$$

Question 3:

Let

$$\alpha = \sqrt{13} + \sqrt{n}$$

a)

$$\begin{aligned}
 \alpha^3 - (n+39)\alpha &= (\sqrt{13} + \sqrt{n})^3 - (n+39)(\sqrt{13} + \sqrt{n}) \\
 &= (\sqrt{13} + \sqrt{n})(\sqrt{13} + \sqrt{n})^2 - (n+39)(\sqrt{13} + \sqrt{n}) \\
 &= (\sqrt{13} + \sqrt{n})(13 + n + 2\sqrt{13}\sqrt{n}) - (n+39)(\sqrt{13} + \sqrt{n}) \\
 &= (\sqrt{13} + \sqrt{n})((13 + n + 2\sqrt{13n}) - (n+39)) \\
 &= (\sqrt{13} + \sqrt{n})(-26 + 2\sqrt{13n}) \\
 &= -26\sqrt{13} + 2 \cdot \sqrt{13}\sqrt{13n} - 26\sqrt{n} + 2\sqrt{13n^2} \\
 &= -26\sqrt{13} + 2 \cdot 13\sqrt{n} - 26\sqrt{n} + 2\sqrt{13n} \\
 &= -16\sqrt{13} + 26\sqrt{n} - 26\sqrt{n} + 2\sqrt{13n} \\
 &= -16\sqrt{13} + 2\sqrt{13n} \\
 &= \sqrt{13}(2n - 26)
 \end{aligned}$$

Hence,

$$\alpha^3 - (n+39)\alpha = \sqrt{13}(2n - 26)$$

and we can write this as,

$$\sqrt{13} = \frac{\alpha^3 - (n+39)}{(2n - 26)}$$

So

$$\sqrt{13} \in \mathbb{Q}_\alpha$$

and since $\sqrt{n} = \alpha - \sqrt{13}$, it follows that

$$\sqrt{n} \in \mathbb{Q}_\alpha$$

Hence,

$$\mathbb{Q}_\alpha = \mathbb{Q}(\sqrt{13}, \sqrt{n})$$

Hence,

$$\mathbb{Q}(\sqrt{13}, \sqrt{n}) = \mathbb{Q}(\sqrt{13} + \sqrt{n})$$

Question 4:

Let

$$\mathbf{F} = \mathbb{Z}_2(\omega)$$

Be the field where

$$\omega^4 = \omega + 1$$